ASSIGNMENT REPORT-6

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For The Subject

Pattern Recognition and Machine Learning

(26AI5707)



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1. INTRODUCTION :

- The Assignment is to generate two classes using Gaussian Mixture Models and classify the data
- We used three Gaussian components for each class and implemented K-Means

2. DATASET :

- The generated dataset contains two classes with three components in each class.
- Total samples : 6,000
- Samples in each class : 3,000
- Training and testing split : 80:20

```
Samples per class: 3000
Components per class: 3
Class 1 weight sum: 1.0
Class 2 weight sum: 1.0
```

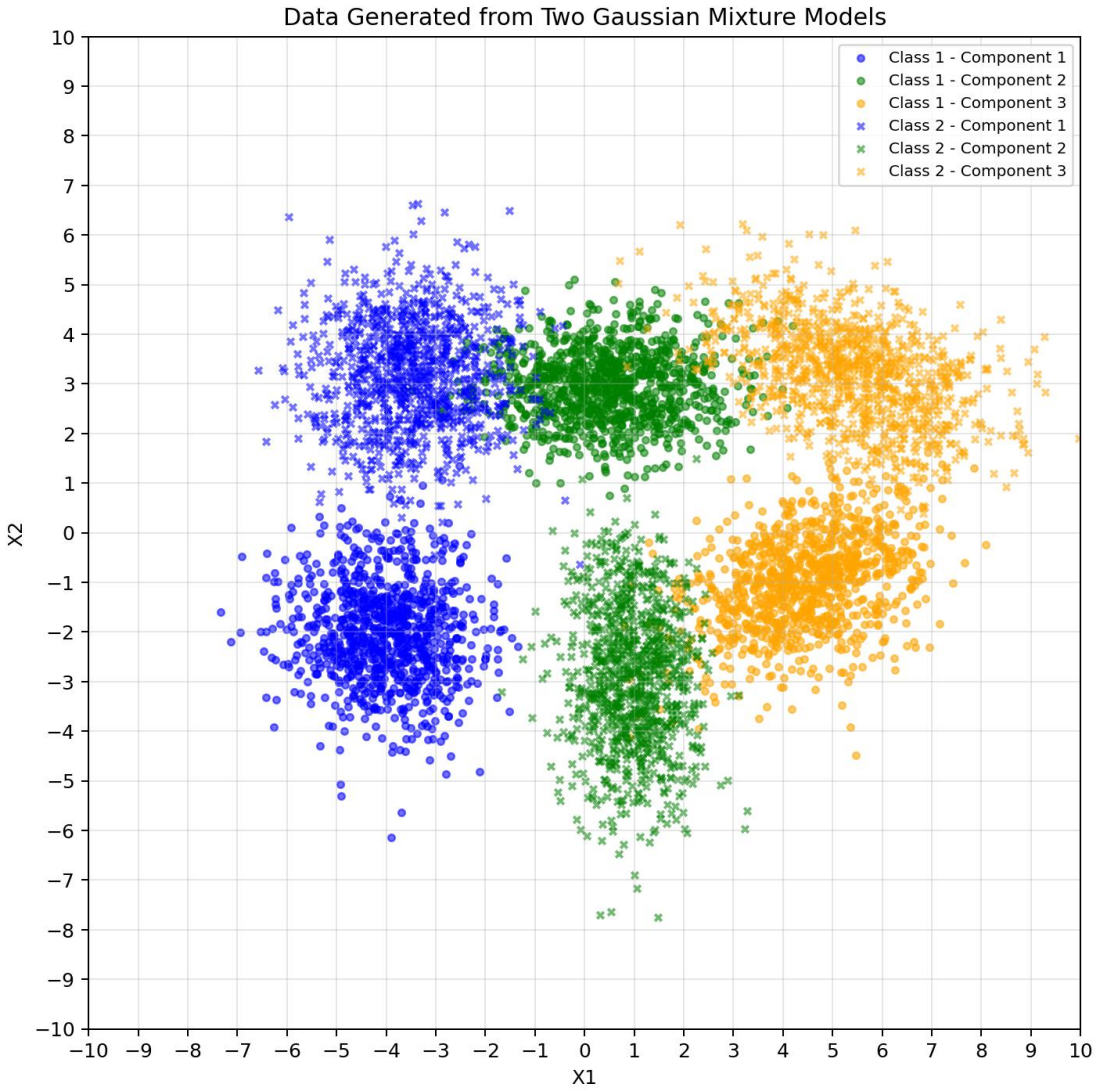


Figure 1 : Generated data for Class 1 and Class 2

3. EXPERIMENT STEPS :

A. GENERATING GAUSSIAN DATA :

- We selected three means, mixture weights and covariance matrices for each class.
- The covariance matrices include spherical, diagonal and full covariance types.
- We used the Box-Muller method to generate standard normal values and then generated the data.

B. DATA SPLITTING :

- We shuffled the data and divided each class into training and testing data in 80:20 ratio.

```
Class 1 training: 2400
Class 1 testing : 600
Class 2 training: 2400
Class 2 testing : 600
```

C. K-MEANS CLUSTERING :

- The nearest centroid was found using squared Euclidean distance.
- The centroids were updated by taking the mean of the points in each cluster.

D. TRAINED CENTROIDS :

```
Class 1 centroids:
[[ 4.449, -0.991]
 [-4.039, -2.029]
 [ 0.443,  2.950]]

Class 2 centroids:
[[ 0.967, -2.912]
 [-3.471,  3.184]
 [ 5.407,  3.197]]
```

E. K-MEANS CLUSTERS AND BOUNDARIES :

- We plotted the training data with the clusters and their centroids.
- The boundary separates the points based on the nearest centroid.

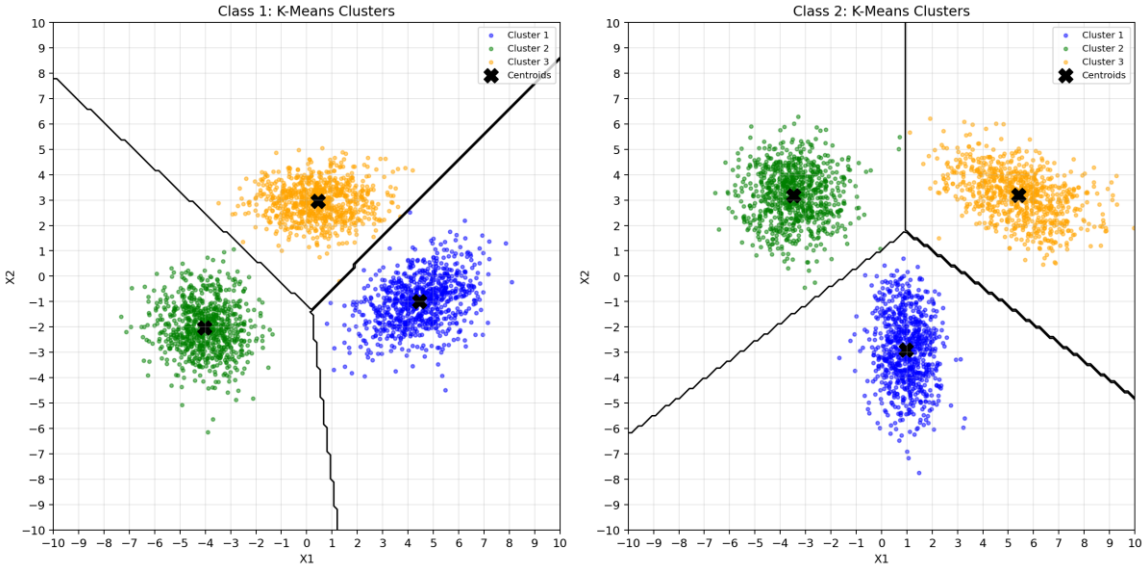


Figure 2 : K-Means clusters and boundaries

F. CLASSIFICATION USING DISTORTION :

- For each testing point, We calculated the minimum squared distance to centroid of each class.
- The point was assigned to the class with the smaller distortion value.

4. RESULTS :

A. CONFUSION MATRIX :

	Predicted Class 1	Predicted Class 2
Actual Class 1	569	31
Actual Class 2	14	586

```
Classification accuracy: 96.25%
Class 1: Precision=0.976, Recall=0.948, F1-score=0.962
Class 2: Precision=0.950, Recall=0.977, F1-score=0.963
```

- Out of 1,200 testing samples, 1,155 were classified correctly.

B. CLASSIFICATION DECISION BOUNDARY :

- The plot shows the class regions based on the minimum distortion.

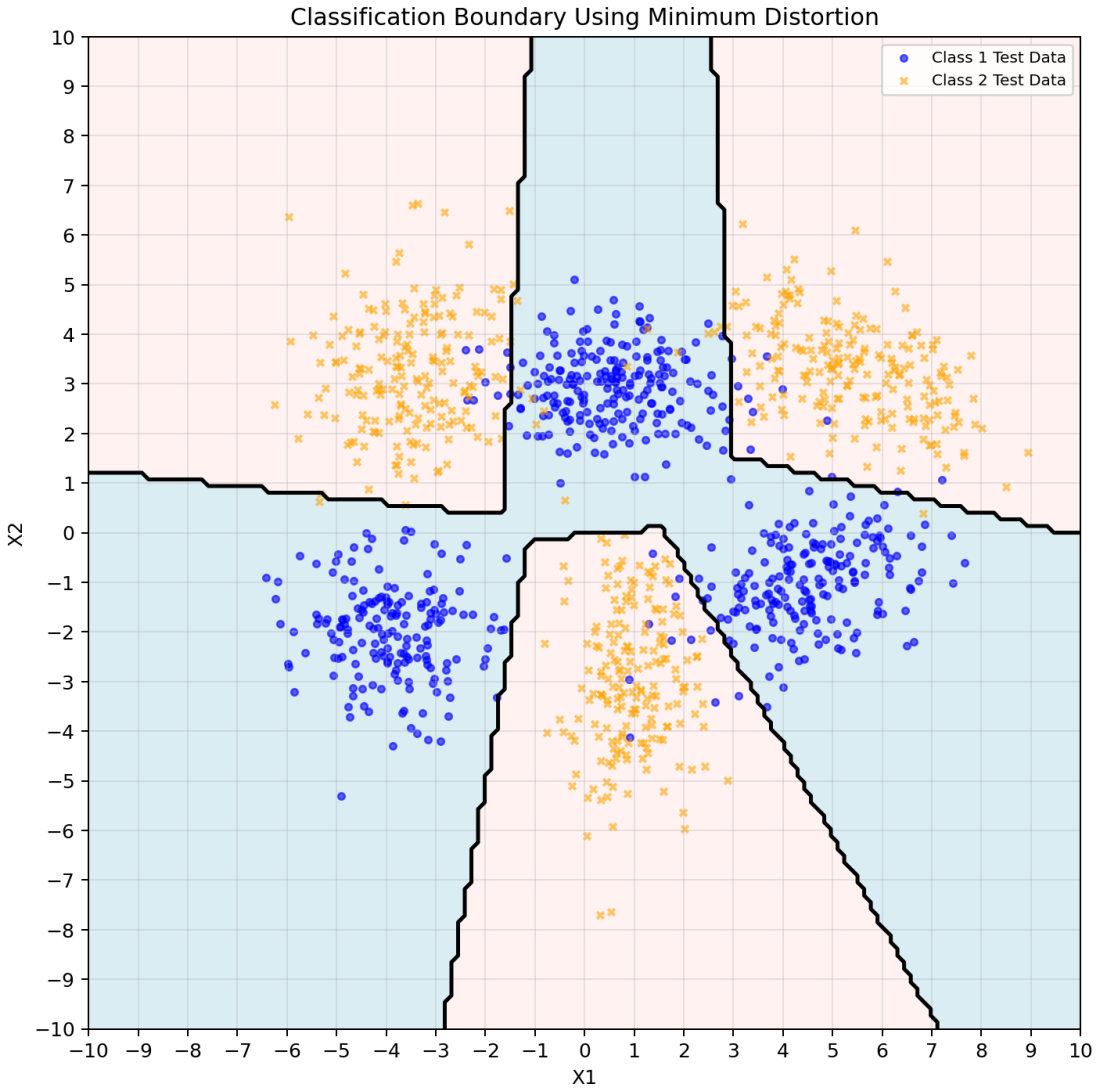


Figure 3 : Classification boundary using minimum distortion

5. INFERENCE :

- K means grouped data around centroids but they not match the original gaussian component centres exactly
- Out of 1200 samples ,1155 were classified correctly and 45 incorrect
- The misclassified points show that two classes overlap
- The classification accuracy obtained in this run was 96.25%.